BSDS Exercise Set 1

Detailed step-by-step solutions in LaTeX

1. Probability generating functions

Full question. Let P, P_1 and P_2 be pgfs.

a) Show that $\lambda P_1 + (1-\lambda)P_2$ is a pgf for any $0 \le \lambda \le 1$. Show that P_1P_2 is a pgf.

b) Is $\exp(P(s)-1)$ a pgf? Fix $0 < \lambda < 1$. Is $P(\lambda s)/P(\lambda)$ necessarily a pgf?

Theory used. A pgf is a power series with nonnegative coefficients summing to 1. The book also notes that a pgf is nondecreasing and convex on $[0,1]$, and that products of pgfs correspond to sums of independent random variables.

Solution.

Let

$$P_1(s) = \sum_{k\ge 0} a_k s^k, \qquad P_2(s) = \sum_{k\ge 0} b_k s^k,$$

where $a_k, b_k \ge 0$ and $\sum_{k\ge 0} a_k = \sum_{k\ge 0} b_k = 1$.

(a) Convex combination. For $0 \le \lambda \le 1$,

$$\lambda P_1(s) + (1-\lambda)P_2(s) = \sum_{k\ge 0} (\lambda a_k + (1-\lambda)b_k)s^k.$$

Each coefficient is nonnegative, and

$$\sum_{k\ge 0} (\lambda a_k + (1-\lambda)b_k) = \lambda \sum_{k\ge 0} a_k + (1-\lambda)\sum_{k\ge 0} b_k = \lambda + (1-\lambda) = 1.$$

Hence $\lambda P_1 + (1-\lambda)P_2$ is a pgf.

Product. Now

$$P_1(s)P_2(s) = \left(\sum_{k\ge 0} a_k s^k\right)\left(\sum_{m\ge 0} b_m s^m\right) = \sum_{n\ge 0} c_n s^n,$$

where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} \ge 0$$

by the Cauchy product. Also,

$$\sum_{n\ge 0} c_n = \left(\sum_{k\ge 0} a_k\right)\left(\sum_{m\ge 0} b_m\right) = 1.$$

Therefore P_1P_2 is a pgf.

(b) The function $\exp(P(s)-1)$. Since $P(s)$ is a pgf, we may write

$$P(s) = \sum_{k\ge 0} p_k s^k, \qquad p_k \ge 0, \quad \sum_{k\ge 0} p_k = 1.$$

Then

$$e^{P(s)-1} = e^{-1}\exp(P(s)) = e^{-1}\sum_{n=0}^{\infty}\frac{P(s)^n}{n!}.$$

For each fixed n, $P(s)^n$ is a pgf because it is the product of n pgfs. The weights $e^{-1}/n!$ are nonnegative and sum to 1:

$$\sum_{n=0}^{\infty}\frac{e^{-1}}{n!} = 1.$$

So $\exp(P(s)-1)$ is a convex combination of pgfs, hence itself a pgf.

The tilted function $\frac{P(\lambda s)}{P(\lambda)}$. Write

$$\frac{P(\lambda s)}{P(\lambda)} = \frac{\sum_{k\geq 0} p_k(\lambda s)^k}{\sum_{k\geq 0} p_k\lambda^k} = \sum_{k\geq 0}\left(\frac{p_k\lambda^k}{P(\lambda)}\right)s^k.$$

All coefficients are nonnegative. Their sum is

$$\sum_{k\geq 0}\frac{p_k\lambda^k}{P(\lambda)} = \frac{P(\lambda)}{P(\lambda)} = 1.$$

Hence, for every $0<\lambda<1$, the function $P(\lambda s)/P(\lambda)$ is also a pgf.

2. Finding pgfs and radii of convergence

Full question. Find the pgf for the following random variables and state their radius of convergence:

a) X where $\mathbb{P}(X=n) = \dfrac{c}{n!}$ for $n\geq 0$. Find the constant c first.

b) Y where $\mathbb{P}(Y=n) = \dfrac{c}{(n+1)(n+2)}$ for $n\geq 0$. Find the constant c first.

c) Z where $\mathbb{P}(Z=n) = \dfrac{1}{2^{n+1}}$ for $n\geq 0$.

Theory used. The book states that a pgf is a power series, is uniquely determined by its coefficients, and has radius of convergence at least 1 for a nonnegative integer-valued random variable.

Solution.

(a)

$$\sum_{n\geq 0}\frac{c}{n!} = c\sum_{n\geq 0}\frac{1}{n!} = ce = 1 \quad\Longrightarrow\quad c = \frac{1}{e}.$$

Hence

$$P_X(s) = \sum_{n\geq 0}\frac{1}{e}\frac{s^n}{n!} = \frac{1}{e}e^s = e^{s-1}.$$

This is an entire function, so the radius of convergence is ∞.

(b) First determine c:

$$\sum_{n\geq 0} \frac{c}{(n+1)(n+2)} = c\sum_{n\geq 0}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = c.$$

Therefore $c = 1$.

Now

$$P_Y(s) = \sum_{n\geq 0} \frac{s^n}{(n+1)(n+2)} = \sum_{n\geq 0}\left(\frac{s^n}{n+1} - \frac{s^n}{n+2}\right).$$

Use the classical series

$$\sum_{n\geq 0} \frac{s^n}{n+1} = -\frac{\ln(1-s)}{s}, \qquad \sum_{n\geq 0} \frac{s^n}{n+2} = \frac{-\ln(1-s) - s}{s^2}.$$

Therefore

$$P_Y(s) = -\frac{\ln(1-s)}{s} - \frac{-\ln(1-s) - s}{s^2} = \frac{s + (1-s)\ln(1-s)}{s^2}.$$

The nearest singularity is at $s = 1$, so the radius of convergence is 1.

(c)

$$P_Z(s) = \sum_{n\geq 0} \frac{s^n}{2^{n+1}} = \frac{1}{2}\sum_{n\geq 0}\left(\frac{s}{2}\right)^n = \frac{1}{2-s}.$$

The radius of convergence is 2.

3. Distribution from a pgf

Full question. Find the distribution of the random variables in each of the cases and find $\mathbb{P}(X \geq n)$ for $n \geq 1$.

a) $P(s) = \dfrac{1}{3-2s}$.

b) $P(s) = \dfrac{1}{4-3s}$.

c) Two dice are rolled. Let X be the sum of the outcomes. Find the pgf of X. Use this pgf to calculate the probability that the sum is divisible by 3.

d) If the pgf of a random variable X is

$$P(s) = \frac{1}{2}e^{s-1} + \frac{1}{4}s^2 + \frac{1}{4},$$

find $\mathbb{P}(X \geq 2)$.

Theory used. The book's pgf examples include geometric distributions, tail probabilities via pgfs, and the product rule for independent sums. The relation $\mathbb{P}(X \geq n) = 1 - \mathbb{P}(X < n)$ is also used.

Solution.

(a)

$$P(s) = \frac{1}{3-2s} = \frac{1}{3}\cdot\frac{1}{1-\frac{2}{3}s} = \sum_{k\geq 0}\frac{1}{3}\left(\frac{2}{3}\right)^k s^k.$$

So

$$\mathbb{P}(X=k) = \frac{1}{3}\left(\frac{2}{3}\right)^k, \qquad k\geq 0,$$

which is a geometric distribution on $\{0,1,2,\dots\}$ with parameter $p=\frac{1}{3}$.

Therefore,

$$\mathbb{P}(X\geq n) = \sum_{k=n}^{\infty}\frac{1}{3}\left(\frac{2}{3}\right)^k = \left(\frac{2}{3}\right)^n, \qquad n\geq 1.$$

(b)

$$P(s) = \frac{1}{4-3s} = \frac{1}{4}\cdot\frac{1}{1-\frac{3}{4}s} = \sum_{k\geq 0}\frac{1}{4}\left(\frac{3}{4}\right)^k s^k.$$

So

$$\mathbb{P}(X=k) = \frac{1}{4}\left(\frac{3}{4}\right)^k, \qquad k\geq 0,$$

a geometric distribution on $\{0,1,2,\dots\}$ with parameter $p=\frac{1}{4}$.

Hence

$$\mathbb{P}(X\geq n) = \left(\frac{3}{4}\right)^n, \qquad n\geq 1.$$

(c) Let U be the outcome of one fair die. Then

$$P_U(s) = \frac{s+s^2+s^3+s^4+s^5+s^6}{6}.$$

Since the two dice are independent, the pgf of their sum is

$$P_X(s) = \left(\frac{s+s^2+s^3+s^4+s^5+s^6}{6}\right)^2.$$

To find the probability that the sum is divisible by 3, use the roots of unity filter. Let

$$\omega = e^{2\pi i/3}, \qquad \omega^2+\omega+1=0.$$

Then

$$\mathbb{P}(X\equiv 0 \pmod 3) = \frac{1}{3}\left(P_X(1)+P_X(\omega)+P_X(\omega^2)\right).$$

Now

$$1+\omega+\omega^2=0 \quad\Longrightarrow\quad 1+\omega+\omega^2+\omega^3+\omega^4+\omega^5=0,$$

so $P_U(\omega)=P_U(\omega^2)=0$. Therefore

$$P_X(\omega)=P_X(\omega^2)=0, \qquad P_X(1)=1,$$

and hence

$$\mathbb{P}(X\equiv 0 \pmod 3) = \frac{1}{3}.$$

(d) *Read the expression as*

$$P(s) = \frac{1}{2}e^{s-1} + \frac{1}{4}s^2 + \frac{1}{4},$$

which is the only pgf-consistent interpretation.

Now

$$P(0) = \frac{1}{2}e^{-1} + \frac{1}{4}, \qquad P'(s) = \frac{1}{2}e^{s-1} + \frac{1}{2}s,$$

so

$$\mathbb{P}(X = 1) = P'(0) = \frac{1}{2}e^{-1}.$$

Therefore

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) = 1 - \left(\frac{1}{2}e^{-1} + \frac{1}{4}\right) - \frac{1}{2}e^{-1} = \frac{3}{4} - \frac{1}{e}.$$

4. Generating functions and convolution

Full question.

a) Use generating functions to prove the combinatorial identity

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

b) Find the sequence whose generating function is $\dfrac{1}{(1-s)^4}$ by expressing it as a product of known generating functions and using convolution.

c) Let a_n be the number of ways to tile a $2 \times n$ board with 1×2 dominoes for $n \geq 1$. Find a recurrence relation for a_n for $n \geq 3$. Set a_1, a_2 appropriately and find its generating function.

Theory used. The book states: "the generating function of $X + Y$ is given by product of the generating function of X and Y" and introduces convolution of sequences.

Solution.

(a) Consider

$$(1+s)^n(1+s)^n = (1+s)^{2n}.$$

The coefficient of s^n on the left is

$$\sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} = \sum_{k=0}^{n} \binom{n}{k}^2.$$

The coefficient of s^n on the right is $\binom{2n}{n}$. Therefore,

$$\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}.$$

(b) We know

$$\frac{1}{(1-s)^2} = \sum_{n \geq 0} (n+1)s^n.$$

Hence

$$\frac{1}{(1-s)^4} = \frac{1}{(1-s)^2} \cdot \frac{1}{(1-s)^2}.$$

By convolution, the coefficient of s^n is

$$c_n = \sum_{k=0}^{n} (k+1)(n-k+1).$$

Now compute:

$$c_n = \sum_{k=0}^{n} ((k+1)(n+1-k)) = \frac{(n+1)(n+2)(n+3)}{6} = \binom{n+3}{3}.$$

So

$$\frac{1}{(1-s)^4} = \sum_{n\geq 0} \binom{n+3}{3} s^n.$$

Thus the sequence is $\left\{\binom{n+3}{3}\right\}_{n\geq 0}$.

(c) A $2 \times n$ board can be tiled in two ways according to the first part:

- place one vertical pair of dominoes in the first column, leaving a $2 \times (n-1)$ board;
- place two horizontal dominoes in the first two columns, leaving a $2 \times (n-2)$ board.

Hence for $n \geq 3$,

$$a_n = a_{n-1} + a_{n-2}.$$

The initial values are

$$a_1 = 1, \qquad a_2 = 2.$$

Let

$$A(s) = \sum_{n\geq 1} a_n s^n.$$

Then

$$A(s) = s + 2s^2 + \sum_{n\geq 3} (a_{n-1} + a_{n-2}) s^n.$$

Now

$$\sum_{n\geq 3} a_{n-1} s^n = s \sum_{m\geq 2} a_m s^m = s\big(A(s) - s\big),$$

and

$$\sum_{n\geq 3} a_{n-2} s^n = s^2 \sum_{m\geq 1} a_m s^m = s^2 A(s).$$

Therefore

$$A(s) = s + 2s^2 + s(A(s) - s) + s^2 A(s).$$

Rearranging,

$$A(s)(1 - s - s^2) = s + s^2,$$

so

$$A(s) = \frac{s + s^2}{1 - s - s^2}.$$

5. First occurrence of patterns and residues modulo 3

Full question.

a) Let p_k be the probability that three consecutive Heads (HHH) is obtained for the first time after the k-th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.

b) Let p_k be the probability of the pattern of two consecutive Heads (HH) or the pattern of Head followed by Tail (HT) is obtained for the first time after the k-th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.

c) Let p_k be the probability that the total obtained in k throws of a dice (not necessarily fair) is divisible by 3. Find the generating function of $\{p_k : k \geq 0\}$. Find $\lim_{k\to\infty} p_k$.

Theory used. The book's generating-function chapter uses first-step recursions and tail/mean identities, and the later chapters use Markov-chain hitting-time recursions. The key idea here is to set up a small state recursion for the waiting time.

Solution.

(a) Three consecutive heads. Let T be the first time the pattern HHH appears. Define

$$F_i(s) = \mathbb{E}[s^T \mid \text{current run of consecutive heads is } i], \qquad i = 0, 1, 2.$$

We need $F_0(s)$.

From state 0:

$$F_0(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_1(s).$$

From state 1:

$$F_1(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_2(s).$$

From state 2:

$$F_2(s) = \frac{s}{2}F_0(s) + \frac{s}{2}.$$

The last equation is because a head completes HHH in one more toss.

Now solve:

$$F_2(s) = \frac{s}{2}(1 + F_0(s)),$$

$$F_1(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_2(s) = \frac{s}{2}F_0(s) + \frac{s^2}{4}(1 + F_0(s)),$$

and then

$$F_0(s) = \frac{s}{2}F_0(s) + \frac{s}{2}F_1(s).$$

Substituting and simplifying gives

$$F_0(s) = \frac{s^3}{8 - 4s - 2s^2 - s^3}.$$

Hence the generating function of $\{p_k\}$ is

$$\sum_{k\geq 0} p_k s^k = \frac{s^3}{8 - 4s - 2s^2 - s^3}.$$

The expected waiting time is

$$\mathbb{E}[T] = F_0'(1) = 14.$$

(b) Pattern HH or HT. The first time either HH or HT appears is exactly one step after the first Head occurs. Indeed, once the first Head is observed, the very next toss completes either HH or HT.

Let N be the time of the first Head. For a fair coin,

$$\mathbb{P}(N = n) = 2^{-n}, \qquad n \geq 1.$$

If T denotes the required time, then $T = N + 1$, so

$$\mathbb{P}(T = k) = 2^{-(k-1)}, \qquad k \geq 2.$$

Therefore

$$\sum_{k\geq 0} p_k s^k = \sum_{k\geq 2} 2^{-(k-1)} s^k = \frac{s^2}{2-s}.$$

The expected time is

$$\mathbb{E}[T] = \sum_{k\geq 2} k\, 2^{-(k-1)} = 3.$$

(c) Sum divisible by 3. Let the probabilities of the die outcomes be $r_1, \ldots, r_6$ and define the residue-class probabilities

$$u_0 = r_3 + r_6, \qquad u_1 = r_1 + r_4, \qquad u_2 = r_2 + r_5.$$

Let $\omega = e^{2\pi i/3}$.

For one throw, the root-of-unity filter gives

$$\alpha := u_0 + u_1\omega + u_2\omega^2, \qquad \overline{\alpha} = u_0 + u_1\omega^2 + u_2\omega.$$

Then for the sum of k independent throws,

$$p_k = \mathbb{P}(S_k \equiv 0 \pmod 3) = \frac{1}{3}(1 + \alpha^k + \overline{\alpha}^k).$$

Therefore the generating function is

$$\sum_{k\geq 0} p_k s^k = \frac{1}{3}\left(\frac{1}{1-s} + \frac{1}{1-\alpha s} + \frac{1}{1-\overline{\alpha} s}\right).$$

For the limit, if the die is not concentrated on a single residue class mod 3, then $|\alpha| < 1$, so

$$\lim_{k\to\infty} p_k = \frac{1}{3}.$$

In the degenerate case where all mass lies in one residue class, the sequence is periodic rather than convergent.

6. Random sums and Poisson thinning

Full question.

a) Claims to an insurance company arrive as a Poisson process with rate λ. The size of each claim is an iid random variable Y with pgf $P_Y(s)$. Let S_n be the total claim amount up to time n. Find the pgf of S_n.

b) A hen lays N eggs, where $N \sim \text{Poisson}(\lambda)$. Each egg hatches with probability p independently. Let K be the number of chicks. Find the pgf of K and identify its distribution.

c) In the previous problem, find the expected number of chicks, $\mathbb{E}[K]$, using the random sum formula and verify by direct computation.

Theory used. The book states: "the pgf of S_N is given by $G(P(s))$" for an independent random sum, and also that the mean of a random sum satisfies $\mathbb{E}(S_N) = \mathbb{E}(N)\mathbb{E}(X_1)$.

Solution.

(a) Let N_n be the number of claims by time n. Since claims arrive according to a Poisson process with rate λ,

$$N_n \sim \text{Poisson}(\lambda n).$$

If the claim sizes are $Y_1, Y_2, \dots$ with common pgf $P_Y(s)$ and

$$S_n = \sum_{i=1}^{N_n} Y_i,$$

then by the random-sum formula,

$$P_{S_n}(s) = \mathbb{E}\Big[P_Y(s)^{N_n}\Big].$$

Because $N_n \sim \text{Poisson}(\lambda n)$, its pgf is

$$G_{N_n}(u) = \exp(\lambda n(u-1)).$$

Hence

$$P_{S_n}(s) = \exp\big(\lambda n(P_Y(s) - 1)\big).$$

(b) Conditional on N, the number of chicks is

$$K \mid N \sim \text{Bin}(N, p).$$

Therefore

$$\mathbb{E}[s^K \mid N] = (1 - p + ps)^N.$$

Taking expectation over $N \sim \text{Poisson}(\lambda)$,

$$P_K(s) = \mathbb{E}[(1-p+ps)^N] = \exp\big(\lambda((1-p+ps)-1)\big) = \exp\big(\lambda p(s-1)\big).$$

This is exactly the pgf of a Poisson(λp) random variable. So

$$K \sim \text{Poisson}(\lambda p).$$

(c) Using the random-sum mean formula,

$$\mathbb{E}[K] = \mathbb{E}[N]\mathbb{E}[\text{chicks from one egg}] = \lambda \cdot p = \lambda p.$$

Directly,

$$\mathbb{E}[K] = \mathbb{E}\big(\mathbb{E}[K \mid N]\big).$$

Given $N = n$, we have $\mathbb{E}[K \mid N = n] = np$, so

$$\mathbb{E}[K] = \sum_{n \geq 0} np\, e^{-\lambda} \frac{\lambda^n}{n!} = p\, e^{-\lambda} \sum_{n \geq 1} n \frac{\lambda^n}{n!}.$$

Now

$$n \frac{\lambda^n}{n!} = \lambda \frac{\lambda^{n-1}}{(n-1)!},$$

hence

$$\mathbb{E}[K] = p\, e^{-\lambda} \lambda \sum_{m \geq 0} \frac{\lambda^m}{m!} = p\, e^{-\lambda} \lambda e^{\lambda} = \lambda p.$$

This agrees with the random-sum formula.